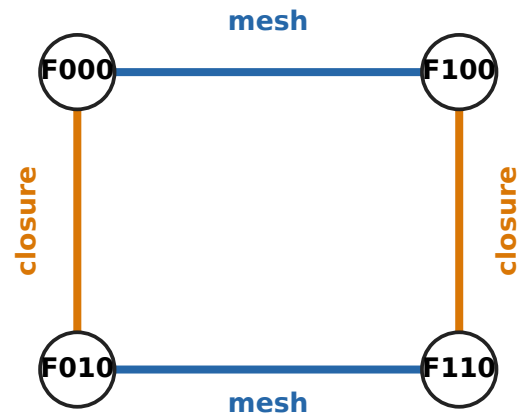


Cost-indexed provenance relaxation in multifidelity UQ

(a) Declared provenance graph



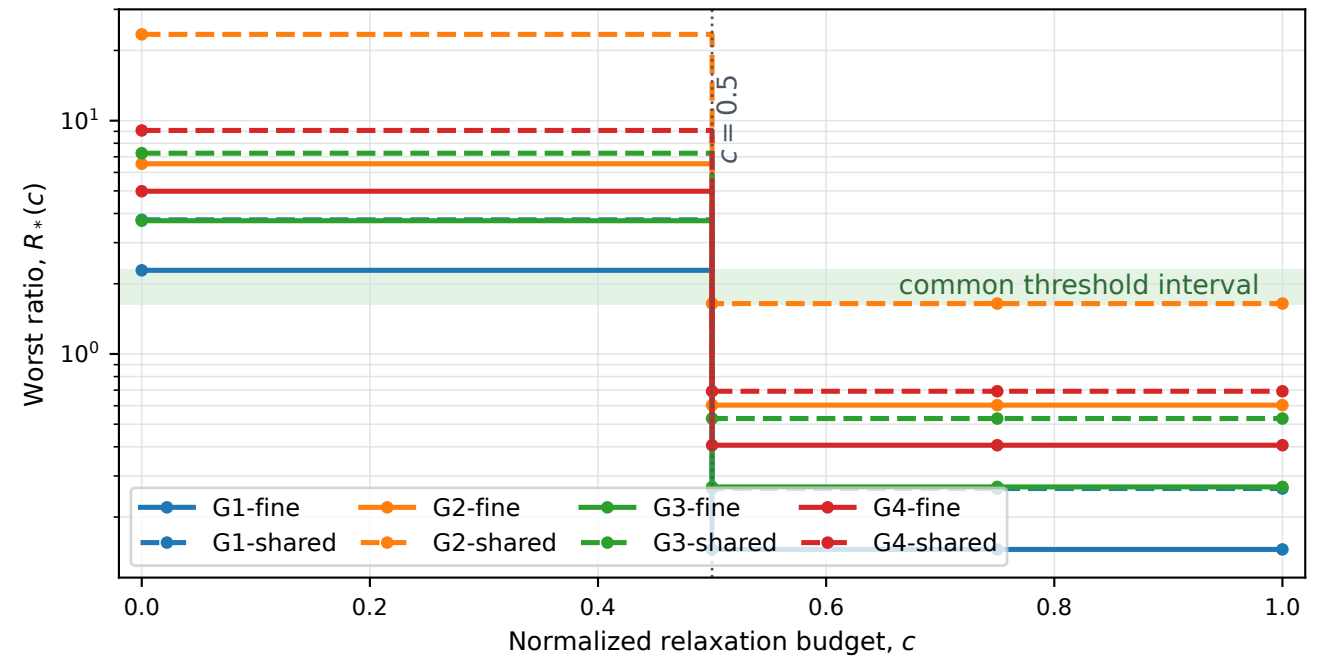
Cut edges permit pathway disagreement

Cost models (total graph cost = 1)

Normalized edge cost	mesh	closure
Balanced	0.25	0.25
Mesh-dominant	0.40	0.10
Closure-dominant	0.10	0.40

Weights are declared sensitivities, not fitted probabilities or physical error magnitudes.

(b) Cost-indexed decision fragility across all eight audits



(c) Exact equality is less stable than near-attainment

